



HarvestWise

Voice-first autonomous restocking for India's street merchants.

WhatsApp / Soundbox



Sarvam Indic Voice AI



Cognee Knowledge Graph



n8n Orchestrator



Mandi Wholesaler

Lakshmi sells vegetables in Bengaluru. Paytm made her a **merchant** — nothing made her a **restocker**. HarvestWise listens, predicts spoilage, recalculates her order and dispatches it to her mandi wholesaler — **without asking her to type a single word.**

Paytm made them merchants. Nobody made them restockers.

India's largest merchant network is also its least-served growth market: payments are digital, but the highest-stakes daily decision — what to buy at 4:30 AM — still runs on guesswork.

30M₊

MERCHANTS IN PAYTM'S ECOSYSTEM

The long tail: kiranas, carts, stalls — the backbone of India's daily commerce.

1.57_{Cr}

MERCHANT DEVICE SUBSCRIPTIONS (SOUNDBOX / POS)

A proven billing rail merchants already pay on — and the anchor for cross-sell.

8_{Cr}

MONTHLY TRANSACTING USERS

Demand-side scale that makes every merchant insight immediately actionable.

The locked layer of value

Paytm sees the transactions — but the merchant's most consequential decision, **what to stock tomorrow at what quantity**, happens blind at the mandi gate. Paytm's own AI (ARMS) optimizes merchant lifecycle *internally*; the merchant's own P&L still runs on intuition. That missing merchant-facing layer is where the next phase of Paytm's growth lives.

Why now: Indic voice AI (Sarvam), agent orchestration (n8n) and knowledge-graph memory (Cognee) matured together for the first time — making a copilot for a non-English, non-dashboard user finally buildable.

Her inventory rots. Her margin goes with it.

"I'm at Kalasipalya mandi by 4:30 AM with ₹3,500 in my pocket. I buy tomatoes, greens and beans on one heuristic: what sold yesterday. At 3 PM an unpredicted thunderstorm empties the street. By 9 PM I'm dumping ₹400 of rotting greens in the garbage. I earn on Paytm. I restock on guesswork."

— LAKSHMI · VEGETABLE CART · BASAVANAGUDI, BENGALURU



Modeled across Bengaluru's 12,000+ vegetable vendors: ~₹175 Cr/yr in spoilage losses (₹400 × 12,000 × 365) — losses like these push vendors into informal high-interest debt (*meter baddi*). Baseline to be validated in pilot.



25–35% daily spoilage

Perishable stock bought at 4:30 AM on intuition, sold across a 12-hour day. Evening leftovers go out at distress prices — what doesn't sell is a 100% cash loss.



4:30 AM <—> 3 PM disconnect

Procurement decisions ignore the signals that actually move demand: afternoon rain, micro-festivals, local UPI purchase velocity, mandi price swings — all available, none connected.



Tools not built for her

Dashboards need literacy, dry hands and downtime she doesn't have. Chatbots only talk. Wet hands, continuous customers, vernacular-only — the interface must be speech.

₹3,500 / day

AVG. WORKING CAPITAL



~30%

UNSOLD AT CLOSE



evening salvage

DISTRESS DISCOUNTING



₹400 / day

NET LOSS · ~₹12,000/MO

Up to 30% of household take-home erased
· illustrative model, spoilage band 25–35%

A restocking loop that speaks, acts & remembers

1 • OBSERVE



Watches her world — hourly UPI sales velocity (Paytm signals, with consent), IMD rain probability, festival calendar, mandi rates, and **perishable decay tables** per commodity.

2 • PREDICT



Forecasts spoilage & demand in her words: *"Rain at 3 PM — coriander shelf-life drops to 4 hours; your rainy-day coriander sales fall 40%."*

3 • ACT



Recalculates and dispatches — resizes her order (10 → 6 bundles), confirms with her by voice, sends the revised buy-order to her mandi wholesaler on WhatsApp.

4 • LEARN



Remembers her business — which recalcs worked, which didn't, and why. Next week's order starts from this week's evidence, not zero.

No new app. No login. No literacy barrier.

The interface is **voice on WhatsApp** — plus the morning briefing **played on her Paytm Soundbox**. The Soundbox already sits on her cart, already speaks her language. We upgrade it from payment confirmation to operational assistant.

Not another chatbot

Chatbots **advise** ("stock less"). HarvestWise **finishes the job** — it recalculates the order and dispatches it to the wholesaler, with a human voice-approval gate before anything that touches money.

One loop, every morning: Observe → Predict → Act → Learn.

Spoilage prevented before it's bought, orders dispatched before the mandi gate, outcomes measured the same evening — compounding daily into her take-home and Paytm's GMV.

A day with HarvestWise — in Kannada

The product story — the finale demo compresses it to one voice command → one explained recalculation → one dispatched order → one remembered outcome.

6:30

AM · BRIEFING

Namaskara Lakshmi! Today: 80% chance of rain at 3 PM. Order fewer tomatoes; leafy greens will sell out by evening. Best mandi rate: ₹18/kg at Kalasipalya.

Auto-sent by the 6:30 AM n8n workflow — voice on WhatsApp **and played on her Soundbox**. 30 seconds, zero taps.

9:14

AM · SHE ORDERS

ಟೋಮ್ಯಾಟೋ 3 ಬಾಕ್ಸ್, ಕೊತ್ತಂಬರಿ 10 ಕಟ್ಟು.
("3 crates of tomatoes, 10 bundles of coriander.")

Order noted. One flag: rain at 3 PM — coriander lasts only 4 hours in humidity, and your rainy-day coriander sales drop ~40%.

Sarvam ASR → intent + entities → Cognition raises the spoilage flag against her history.

11:02

AM · IT RECALCULATES

Revised: 6 bundles of coriander (not 10), 3 crates tomatoes. Wholesaler Ravi anna — ₹546. Approve?

ಹೌದು, ಕಳುಹಿಸು. ("Yes, send it.")

Dispatched: buy-order sent to the wholesaler, stock sheet updated — n8n gate: nothing moves without her "yes".

8:30

PM · IT LEARNS

Today: sold out by 7:15 PM, zero coriander waste. You kept ₹1,940 you'd have discounted away. This week: +₹4,100 vs last.

Logged to her Cognition memory graph: **"Rain recalc — worked, ₹1,940 kept."** Next storm, she doesn't have to think.

Built on the organizers' own stack — one job each

Experience

WHERE THE MERCHANT LIVES

WhatsApp Business API

voice notes, in & out

Paytm Soundbox

morning audio briefing

Business with Paytm

optional dashboard view

No new app, no login, no typing. Voice is the entire UI; the Soundbox is the loudspeaker.

Intelligence

SARVAM AI · INDIC SPEECH LAYER

Sarvam Speech-to-Text

ASR · 22+ Indic languages

Sarvam LLM

intent + entities + reasoning

Sarvam Text-to-Speech

natural voice · 11 languages

Indic-native understanding — not an English product with translation bolted on.

Decision Core

COGNEE · CAUSAL KNOWLEDGE GRAPHS

Merchant memory graph

her history · what worked

Commodity decay tables

shelf-life × humidity × rain

Cold-start patterns

anonymized cross-merchant

Shelf-life × rain × her UPI velocity — the causal layer behind every recalc.

Orchestration

N8N · EVENT-DRIVEN ACTIONS

n8n 6:30 AM workflow

briefing · WhatsApp + Soundbox

n8n order dispatcher

revised buy-order → wholesaler

n8n approval gates

no money action without "yes"

Event-driven dispatch to mandi wholesalers; escalation & fallback built in.

Data

CONSENT-FIRST SIGNALS

Paytm merchant signals

hourly UPI velocity · consent

IMD weather API

afternoon rain probability

Mandi rate feeds

public daily price feeds

Demo runs on Paytm-transaction-shaped seeded data (labeled as such); production stays consent-first inside Paytm's rails.

■ **GUARDRAILS ARE THE PRODUCT** DPDP-aligned · federated learning (patterns shared, raw data never) · human approval on every money action · every recalc explained & undoable · full audit trail

Advice is cheap. Closed loops are rare.

Every alternative stops at a screen or a sentence. HarvestWise is the only one that speaks her language, recalculates her order, dispatches it, and remembers her business.

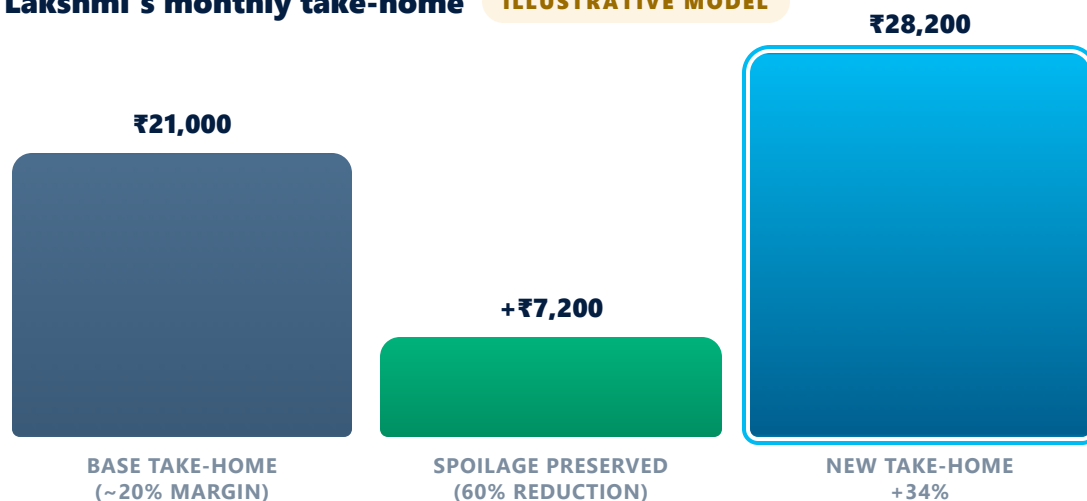
Evaluation	Traditional merchant tools	Generic AI chatbots	HarvestWise
Input modality	English forms & dashboards — needs literacy, dry hands, downtime	Text-only chat — assumes she can type	Zero-touch Indic voice notes on WhatsApp (Sarvam) — she talks; that's the whole UI
Decision intelligence	Static inventory tallying — backwards-looking	Generic public advice — no context, no memory	Causal knowledge graph: shelf-life × rain × her hourly UPI velocity (Cognee)
Workflow completion	Manual execution — she still does the work	Advises only — "stock less", then stops	Closed-loop procurement: recalculated buy-order dispatched to her mandi wholesaler (n8n)
Hardware tie-in	None	None	Morning audio briefing on Paytm Soundbox — payment speaker upgraded to operational assistant

MOAT Per-merchant memory graph (Cognee) × vernacular closed loop (Sarvam + n8n) × Soundbox/Paytm-native distribution — a compounding data advantage every single day.

Waste is her biggest expense. We give 60% of it back.

Lakshmi's monthly take-home

ILLUSTRATIVE MODEL



Assumes ₹3,500 avg. daily working capital × 30 selling days, ~20% gross margin, ₹400/day net spoilage loss after evening salvage. The 60% spoilage-reduction target (₹240/day ≈ ₹7,200/mo) is the pilot metric we commit to validating — not a claim.

What Paytm wins

- Soundbox upgraded:** from passive payment-confirmation speaker to **active operational assistant** — a deeper, stickier device relationship.
- GMV up:** better-stocked merchants capture demand they currently miss — target **+8% GMV per merchant**.
- Underwriting data moat:** proprietary inventory & procurement telemetry derisks **Paytm Merchant Loans** — replacing "meter baddi" with formal credit.
- Mission delivered:** financial inclusion that shows up as a healthier family P&L — the story of Build for India.

₹7,200

PRESERVED / VENDOR / MO*

₹105 Cr+

WORKING CAPITAL / YR · 12K VENDORS*

₹175 Cr+

CURRENT ANNUAL SPOILAGE LOSS (CITY-WIDE EST.)

*Model: 12,000 vendors × ₹240/day × 365 — directional; validated in pilot. Impact framed as money preserved, not vanity metrics.

Billed where she already pays. Free where she starts.

Seed

₹0 / forever

- **Daily demand alerts** — weather + spoilage risk, one voice note + Soundbox briefing
- Yesterday's sales recap
- Builds the daily habit — **zero CAC** via Paytm for Business & Soundbox base

SETTLEMENT-BILLED

Pro

₹149 / month

- **Autonomous restocking:** order recalculation + dispatch to her wholesaler
- Price-drop supplier alerts · end-of-day **P&L voice audit**
- Proposed billing: **settlement-linked micro-deduction** — subject to Paytm integration approval; **simulated in our prototype**. No invoice, no friction.

Platform Revenues

Take-rate on value created

- **Supplier marketplace:** % on executed mandi orders
- **Credit distribution:** procurement telemetry derisks Paytm Merchant Loans
- **Anonymized demand curves** (DPDP-compliant) for supply-chain aggregators & mandi logistics planning

Why merchants buy

₹149/mo vs ₹7,200/mo preserved — pays for itself in one saved evening of stock.

Why it compounds

Every day improves her memory graph — advice quality compounds, churn falls.

Unit economics

Zero-CAC Paytm-native distribution × 80%+ SaaS margin × data flywheel into lending.

Why Paytm wins

The deepest merchant lock-in since the Soundbox: the device confirmed payments; we grow them.

THE WEDGE → THE PLATFORM Start: perishable carts & stalls (highest pain, clearest signal) → kirana fresh categories → 30M+ merchants as the daily restocking layer of the Paytm ecosystem.

[NEXT](#)[ROADMAP](#) · [TEAM](#) · [ASK](#)

From this deck to a pilot

NOW — FINALE

MVP on partner stack

Synthetic 90-day sales data; 3 n8n workflows (briefing, order recalc + dispatch, day-wrap learning); Sarvam voice loop; Cognee memory. Live demo, one phone.

MONTHS 1-3

500-merchant pilot

KR Market, Russell Market & Madivala with vendor associations + Paytm's merchant network. Validate the 60% spoilage-reduction metric.

MONTHS 4-6

Paytm-native integration

API partnership with Paytm for Business; Soundbox briefing rail; mandi logistics channels wired into n8n dispatch.

BEYOND

The restocking layer

Perishables → kirana fresh → all categories. Procurement telemetry → merchant credit. Every merchant, a copilot.

"Every morning at 4:30, India's merchants guess. From tomorrow, they guess **with a copilot.**"

TEAM

Sairam — Venkata Narayana G V · Malaravan P

BUILT WITH

Sarvam AI · n8n · Cognee

ASK

A 500-merchant pilot with Paytm's network

Solve with AI. Build for India.